EXERCISE 1 Find the equation of the tangent to :

1. $y = x - 2x^2 + 3$ at $x = 2$
2. $y = \sqrt{x} + 1$ at $x = 4$
3. $y = x^3 - 5x$ at $x = 1$
4. $y = \dfrac{4}{\sqrt{x}}$ at $(1, 4)$
5. $y = \dfrac{3}{x} - \dfrac{1}{x^2}$ at $(-1, -4)$
6. $y = 3x^2 - \dfrac{1}{x}$ at $x = -1$

Check your answers using technology.

EXERCISE 2 Find the equation of the normal to $y = \dfrac{8}{\sqrt{x}}$ at the point where $x = 4$.

EXERCISE 3 Find the equation of the normal to :

1. $y = x^2$ at the point $(3, 9)$
2. $y = x^3 - 5x + 2$ at $x = -2$
3. $y = \dfrac{5}{\sqrt{x}} - \sqrt{x}$ at the point $(1, 4)$
4. $y = 8\sqrt{x} - \dfrac{1}{x^2}$ at $x = 1$

EXERCISE 4 Find the equations of any horizontal tangents to

$$y = x^3 - 12x + 2.$$

EXERCISE 5 Find the equations of the horizontal tangents to

$$y = 2x^3 + 3x^2 - 12x + 1.$$

EXERCISE 6 Find the points of contact where horizontal tangents meet the curve

$$y = 2\sqrt{x} + \frac{1}{\sqrt{x}}.$$

EXERCISE 7 Find k if the tangent to

$$y = 2x^3 + kx^2 - 3$$

at the point where $x = 2$ has gradient 4.

EXERCISE 8 Find the equation of the other tangent to

$$y = 1 - 3x + 12x^2 - 8x^3$$

which is parallel to the tangent at $(1, 2)$.

EXERCISE 9

1. Consider the curve $y = x^2 + ax + b$ where a and b are constants. The tangent to this curve at the point where $x = 1$ is $2x + y = 6$. Find the values of a and b.
2. Consider the curve $y = a\sqrt{x} + \dfrac{b}{\sqrt{x}}$ where a and b are constants. The normal to this curve at the point where $x = 4$ is $4x + y = 22$. Find the values of a and b.
3. Show that the equation of the tangent to $y = 2x^2 - 1$ at the point where $x = a$, is

$$4ax - y = 2a^2 + 1.$$

EXERCISE 10 Find the equation of the tangent to :

1. $y = \sqrt{2x+1}$ at $x = 4$
2. $y = \dfrac{1}{2-x}$ at $x = -1$
3. $f(x) = \dfrac{x}{1-3x}$ at $(-1, -\frac{1}{4})$
4. $f(x) = \dfrac{x^2}{1-x}$ at $(2, -4)$

EXERCISE 11 Find the equation of the normal to :

1. $y = \dfrac{1}{(x^2+1)^2}$ at $(1, \frac{1}{4})$
2. $y = \dfrac{1}{\sqrt{3-2x}}$ at $x = -3$
3. $f(x) = \sqrt{x}(1-x)^2$ at $x = 4$
4. $f(x) = \dfrac{x^2-1}{2x+3}$ at $x = -1$

EXERCISE 12 Find the equation of :

1. the tangent to the function $f : x \mapsto e^{-x}$ at the point where $x = 1$
2. the tangent to $y = \ln(2-x)$ at the point where $x = -1$
3. the normal to $y = \ln(\sqrt{x})$ at the point where $y = -1$

EXERCISE 13 Find the equation of the tangent to $y = \tan x$ at the point where $x = \frac{\pi}{4}$.

EXERCISE 14 Find the equation of :

1. the tangent to $y = \sin x$ at the origin.
2. the tangent to $y = \tan x$ at the origin.
3. the normal to $y = \cos x$ at the point where $x = \dfrac{\pi}{6}$.
4. the normal to $y = \dfrac{1}{\sin(2x)}$ at the point $x = \dfrac{\pi}{4}$.

EXERCISE 15

1. Find where the tangent to the curve $y = x^3$ at the point where $x = 2$, meets the curve again.
2. Find where the tangent to the curve $y = -x^3 + 2x^2 + 1$ at the point where $x = -1$, meets the curve again.

EXERCISE 16 Consider the function $f(x) = x^2 + \frac{4}{x^2}$.

1. Find $f'(x)$.
2. Find the values of x at which the tangent to the curve is horizontal.
3. Show that the tangents at these points are the same line.

EXERCISE 17 The tangent to $y = x^2e^x$ at $x = 1$ cuts the x- and y-axes at A and B respectively.

Find the coordinates of A and B.

EXERCISE 18

1. Find the equation of the tangent to

$$y = x^2 - x + 9$$

at the point where $x = a$. Hence, find the equations of the two tangents from $(0, 0)$ to the curve. State the coordinates of the points of contact.

2. Find the equations of the tangents to $y = x^3$ from the external point $(-2, 0)$.
3. Find the equation of the normal to $y = \sqrt{x}$ from the external point $(4, 0)$.

Hint : There is no normal at the point where $x = 0$, as this is the endpoint of the function.

EXERCISE 19 Consider the function $f(x) = \frac{8}{x^2}$.

1. Sketch the graph of the function.
2. Find the equation of the tangent at the point where $x = a$.
3. If the tangent in **(b)** cuts the x-axis at A and the y-axis at B, find the coordinates of A and B.
4. Find the area of triangle OAB and discuss the area of the triangle as $a \to \infty$.

EXERCISE 20 Consider $f(x) = \dfrac{2x-3}{x^2+2x-3}$.

1. Show that

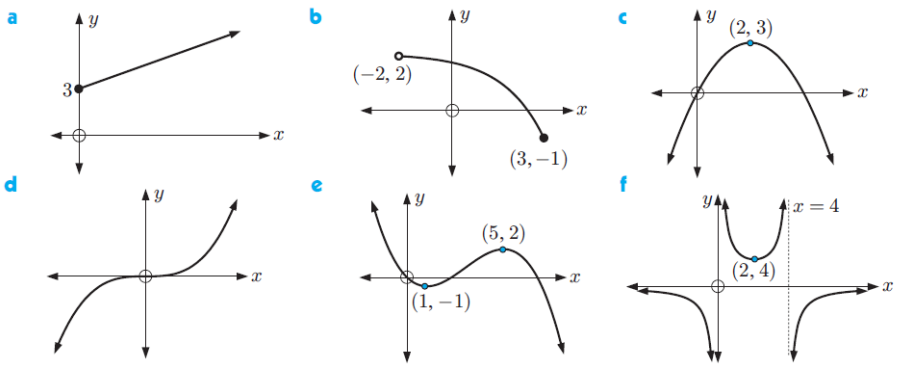
$$f'(x) = \frac{-2x(x-3)}{(x+3)^2(x-1)^2}$$

and draw its sign diagram.

2. Hence, find the intervals where $y = f(x)$ is increasing or decreasing.

EXERCISE 21 Write down the intervals where the graphs are :

1. increasing
2. decreasing



EXERCISE 22 Find the intervals where $f(x)$ is increasing or decreasing :

1. $f(x) = x^2$
2. $f(x) = -x^3$
3. $f(x) = 2x^2 + 3x - 4$
4. $f(x) = \sqrt{x}$
5. $f(x) = \dfrac{2}{\sqrt{x}}$
6. $f(x) = x^3 - 6x^2$
7. $f(x) = e^x$
8. $f(x) = \ln x$
9. $f(x) = -2x^3 + 4x$
10. $f(x) = -4x^3 + 15x^2 + 18x + 3$
11. $f(x) = 3 + e^{-x}$
12. $f(x) = xe^x$
13. $f(x) = 3x^4 - 16x^3 + 24x^2 - 2$
14. $f(x) = 2x^3 + 9x^2 + 6x - 7$
15. $f(x) = x^3 - 6x^2 + 3x - 1$
16. $f(x) = x - 2\sqrt{x}$

EXERCISE 23 Consider $f(x) = \dfrac{4x}{x^2+1}$.

1. Show that
$$f'(x) = \frac{-4(x+1)(x-1)}{(x^2+1)^2}$$
and draw its sign diagram.
2. Hence, find intervals where $y = f(x)$ is increasing or decreasing.

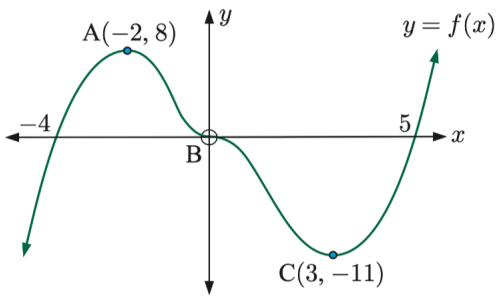
EXERCISE 24 Consider $f(x) = \dfrac{4x}{(x-1)^2}$.

1. Show that
$$f'(x) = \frac{-4(x+1)}{(x-1)^3}$$
and draw its sign diagram.
2. Hence, find intervals where $y = f(x)$ is increasing or decreasing.

EXERCISE 25 Consider $f(x) = \dfrac{-x^2+4x-7}{x-1}$.

1. Show that
$$f'(x) = \frac{-(x+1)(x-3)}{(x-1)^2}$$
and draw its sign diagram.
2. Hence, find intervals where $y = f(x)$ is increasing or decreasing.

EXERCISE 26 The tangents at points A, B and C are horizontal.



1. Classify points A, B and C.
2. Draw a sign diagram for the gradient function $f'(x)$ for all x.
3. State intervals where $y = f(x)$ is :
   a. increasing
   b. decreasing
4. Draw a sign diagram for $f(x)$ for all x.

EXERCISE 27 For each of the following functions, find and classify any stationary points. Sketch the function, showing all important features.

1. $f(x) = x^2 - 2$
2. $f(x) = x^3 + 1$
3. $f(x) = x^3 - 3x + 2$
4. $f(x) = x^4 - 2x^2$
5. $f(x) = x^3 - 6x^2 + 12x + 1$
6. $f(x) = \sqrt{x} + 2$
7. $f(x) = x - \sqrt{x}$
8. $f(x) = x^4 - 6x^2 + 8x - 3$
9. $f(x) = 1 - x\sqrt{x}$
10. $f(x) = x^4 - 2x^2 - 8$

EXERCISE 28 At what value of x does the quadratic function $f(x) = ax^2 + bx + c,\ a \neq 0$, have a stationary point ? Under what conditions is the stationary point a local maximum or a local minimum ?

EXERCISE 29 Find the position and nature of the stationary point(s) of :

1. $y = xe^{-x}$
2. $y = x^2e^x$
3. $y = \dfrac{e^x}{x}$
4. $y = e^{-x}(x + 2)$

EXERCISE 30 $f(x) = 2x^3 + ax^2 - 24x + 1$ has a local maximum at $x = -4$. Find a.

EXERCISE 31 $f(x) = x^3 + ax + b$ has a stationary point at $(-2, 3)$.

1. Find the values of a and b.
2. Find the position and nature of all stationary points.

EXERCISE 32 Consider $f(x) = x \ln x$.

1. For what values of x is $f(x)$ defined ?
2. Show that the global minimum value of $f(x)$ is $-\dfrac{1}{e}$.

EXERCISE 33 Prove that

$$\frac{\ln x}{x} \leq \frac{1}{e} \quad \text{for all } x > 0.$$

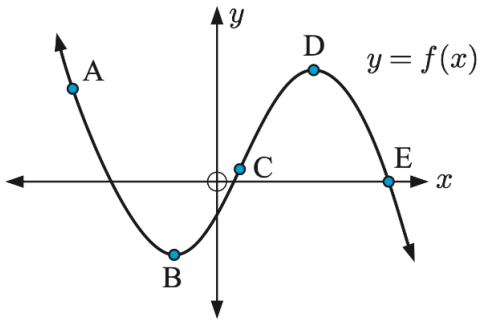
Hint : Let $f(x) = \frac{\ln x}{x}$ and find its greatest value.

EXERCISE 34 Consider the function $f(x) = x - \ln x$.

1. Show that the graph of $y = f(x)$ has a local minimum and that this is the only turning point.
2. Hence prove that $\ln x \leq x - 1$ for all $x > 0$.

EXERCISE 35

1. In the diagram shown, B and D are stationary points, and C is a point of inflection. Complete the table by indicating whether each value is zero, positive, or negative :



Point	$f(x)$	$f'(x)$	$f''(x)$
A	$+$		
B		0	
C			
D		0	
E			

2. Describe the turning points of $y = f(x)$.
3. Describe the inflection point of $y = f(x)$.

EXERCISE 36 Find and classify all points of inflection of :

1. $f(x) = x^2 + 3$
2. $f(x) = 2 - x^3$
3. $f(x) = x^3 - 6x^2 + 9x + 1$
4. $f(x) = x^3 + 6x^2 + 12x + 5$
5. $f(x) = -3x^4 - 8x^3 + 2$
6. $f(x) = 3 - \frac{1}{\sqrt{x}}$

EXERCISE 37 For each of the following functions :

1. **a.** Find and classify all points where $f'(x) = 0$
   b. Find and classify all points of inflection
   c. Find intervals where the function is increasing or decreasing
   d. Find intervals where the function is concave up or down
   e. Sketch the function showing the features you have found.

2. **a.** $f(x) = x^2$

b. $f(x) = x^3$

c. $f(x) = \sqrt{x}$

d. $f(x) = x^3 - 3x^2 - 24x + 1$

e. $f(x) = 3x^4 + 4x^3 - 2$

f. $f(x) = (x-1)^4$

g. $f(x) = x^4 - 4x^2 + 3$

h. $f(x) = 3 - \dfrac{4}{\sqrt{x}}$

EXERCISE 38 The function $f(x) = e^{2x} - 3$ cuts the x-axis at A and the y-axis at B.

1. Find the coordinates of A and B.

2. Show algebraically that the function is increasing for all x.

3. Find $f''(x)$ and hence explain why $f(x)$ is concave up for all x.

4. Use technology to help graph $y = e^{2x} - 3$.

5. Explain why $y = -3$ is a horizontal asymptote.

EXERCISE 39 Suppose $f(x) = e^x - 3$ and $g(x) = 3 - 5e^{-x}$ where $-3 \leq x \leq 4$.

1. Find the x- and y-intercepts of both functions.

2. Discuss $f(x)$ and $g(x)$ as $x \to \infty$ and as $x \to -\infty$.

3. Find algebraically the point(s) of intersection of the functions.

4. Sketch the graph of both functions on the same set of axes. Show all important features on your graph.

EXERCISE 40 The function $y = e^x - 3e^{-x}$ cuts the x-axis at P and the y-axis at Q.

1. Determine the coordinates of P and Q.

2. Prove that the function is increasing for all x.

3. Show that $\dfrac{d^2y}{dx^2} = y$. What can be deduced about the concavity of the function above and below the x-axis ?

4. Use technology to help graph $y = e^x - 3e^{-x}$. Show the features of **a**, **b**, and **c** on the graph.

EXERCISE 41 Consider $f(x) = \ln(2x-1) - 3$.

1. Find the x-intercept.
2. Can $f(0)$ be found ? What is the significance of this result ?
3. Find the gradient of the tangent to the curve at $x = 1$.
4. Find the domain of f.
5. Find $f''(x)$ and hence explain why $f(x)$ is concave down for all x in the domain of f.
6. Graph the function, showing the features you have found.

EXERCISE 42 Consider the function $f(x) = \dfrac{e^x}{x}$.

1. Does the graph of $y = f(x)$ have any x- or y-intercepts ?
2. Discuss $f(x)$ as $x \to \infty$ and as $x \to -\infty$.
3. Find and classify any stationary points of $y = f(x)$.
4. Find the intervals where $f(x)$ is : **i)** concave up **ii)** concave down.
5. Sketch the graph of $y = f(x)$ showing all important features.
6. Find the equation of the tangent to $f(x) = \dfrac{e^x}{x}$ at the point where $x = -1$.

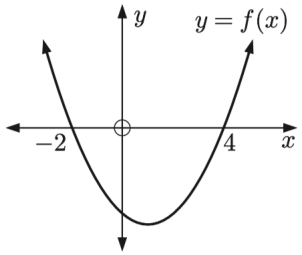
EXERCISE 43 A function commonly used in statistics is the *normal distribution function*

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}.$$

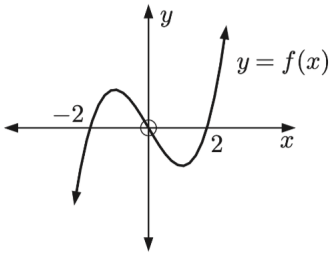
1. Find the stationary points of the function and find the intervals where the function is increasing and decreasing.
2. Find all points of inflection.
3. Discuss $f(x)$ as $x \to \infty$ and as $x \to -\infty$.
4. Sketch the graph of $y = f(x)$ showing all important features.

EXERCISE 44 Using the graphs of $y = f(x)$ below, sketch the graphs of $y = f'(x)$ and $y = f''(x)$. Show clearly the axes intercepts and turning points.

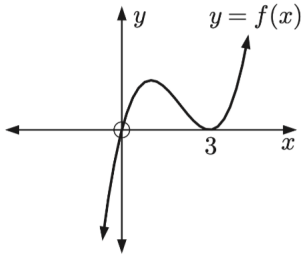
a



b



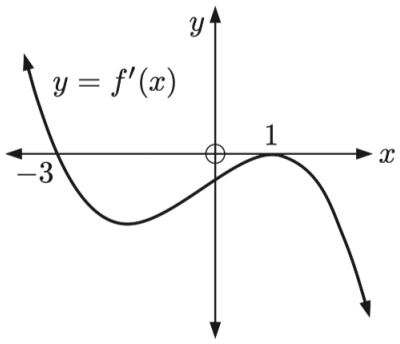
c



EXERCISE 45

For the graphs of $y = f'(x)$ below, sketch a graph which could be $y = f(x)$. Show clearly the location of any stationary points and points of inflection.

a



b

